Connecting Goal Semantics to Learning Geometry: Distinctions, Symmetry Breaking, and Singularity Types

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Abstract

How can we detect when an AI agent's goals have semantically changed? Singular learning theory (SLT) offers geometric diagnostics, but geometry is not semantics: goals can drift without obvious changes in complexity measures. This paper develops a mechanism connecting the two.

The key insight: *semantics are implemented through distinctions*. Distinctions induce partitions; partitions determine symmetries; symmetries cause singularities. We prove a template theorem showing that coarsening distinctions monotonically enlarges symmetry groups, while refining distinctions breaks symmetries. Combined with standard SLT results linking symmetries to singularity structure, this provides a concrete pathway from semantic change to detectable geometric change.

We formalize the "faithfulness" assumption (distinct semantic equilibria yield non-equivalent singularity types), identify sufficient conditions via the distinction-symmetry mechanism, and instantiate the template in three AI architectures: causal-coding networks, incremental compression, and probabilistic logic.

Contents

1 Introduction: The Gap Between Geometry and Semantics

1.1 The goal stability problem

Consider an AI agent capable of modifying its own objectives or representations. A central question in AI safety is: *when do such modifications preserve the semantic content of the agent's goals, and when do they constitute genuine goal drift?*

Singular learning theory (SLT) offers a promising approach: monitor geometric invariants of the learning landscape—the real log canonical threshold (RLCT), local learning coefficients, or related quantities—and use changes in these diagnostics to detect structural shifts. We assume basic familiarity with SLT; recall that the RLCT λ controls the leading term in the asymptotics of the Bayesian marginal likelihood (Watanabe 2009), and that singularities arise from non-identifiabilities in the model parameterization.

However, there is a fundamental gap: **geometry is not semantics**. Many distinct representational states can share similar global diagnostics. An agent might drift from "maximize human flourishing" to "maximize reported satisfaction scores" while the coarse geometric signature of its loss landscape remains unchanged.

1.2 The faithfulness question

To bridge this gap, one needs a stronger structural assumption:

> *Distinct semantic goal equilibria correspond to non-equivalent singularity types.*

If this "faithfulness" condition holds, then detecting a change in singularity type implies a semantic change. But when is such an assumption justified?

1.3 This paper's contribution

We provide a concrete mechanism linking semantics to singularities, built on a simple observation: *semantics are implemented through distinctions*—what the agent treats as equivalent vs. different. Distinctions induce partitions; partitions determine symmetry groups; symmetries cause singularities.

Our main technical contribution is a **template theorem** (Theorem 4.5) showing that coarsening distinctions *always* enlarges the symmetry group and increases redundancy, while refining distinctions *always* shrinks symmetries. Combined with standard results connecting symmetries to RLCT (Example 2.2), this provides a falsifiable mechanism by which semantic changes propagate to geometric changes.

This paper addresses *detection* of semantic changes via geometry, not *prevention* of goal drift. Even with perfect detection, additional mechanisms would be needed to halt or reverse drift.

1.4 Paper structure

Section 2 provides conceptual overview with a concrete toy example. Section 3 formalizes semantic states and the faithfulness assumption, including the explicit map from semantics to partitions. Section 4 develops the partition template theorem. Section 5 gives three instantiations. Section 6 discusses implications and limitations.

2 Conceptual Overview

2.1 The distinction-symmetry-singularity chain

The mechanism linking semantics to geometry proceeds in steps:

Step 1: Semantics as distinctions. A semantic state determines which situations the agent treats as equivalent (same goal-relevant features) vs. different. A paperclip maximizer distinguishes by paperclip count; a welfare maximizer distinguishes by wellbeing indicators.

Step 2: Distinctions induce partitions. Given a distinction pattern, we obtain a partition: items treated as equivalent fall in the same block. Coarser distinctions yield coarser partitions (fewer, larger blocks).

Step 3: Partitions determine symmetries. The automorphism group $\mathrm{Aut}(\Pi)$ consists of permutations preserving block membership. Coarser partitions have larger automorphism groups: merging blocks of sizes j and k replaces $\mathrm{Sym}(j) \times \mathrm{Sym}(k)$ with $\mathrm{Sym}(j + k)$.

Step 4: Symmetries cause singularities. In learning, symmetries manifest as non-identifiabilities: multiple parameter settings achieving the same loss. This creates flat directions, non-isolated minima, and degenerate Fisher information—precisely what SLT's singularity analysis captures.

2.2 A toy example: the full chain in action

Before proceeding to formal development, we illustrate the complete chain with a minimal example.

Example 2.1 (Two-hidden-unit network). Consider a neural network $f(x; w_1, w_2) = \sigma(w_1 x) + \sigma(w_2 x)$ with two hidden units and weights $w_1, w_2 \in \mathbb{R}$.

Semantic state: Suppose the agent's "semantics" are defined by which input regions it distinguishes. If the agent treats inputs symmetrically (no distinction between "w_1-relevant" and "w_2-relevant" structure), the two hidden units are semantically interchangeable.

Partition: The parameter space is $\Theta = \mathbb{R}^2$. Under the symmetric semantics, the partition on parameters is $\Pi_{\text{coarse}} = \{\{w_1, w_2\}\}$—both weights in one block (interchangeable).

Symmetry: $\text{Aut}(\Pi_{\text{coarse}}) = \text{Sym}(\{w_1, w_2\}) \cong \mathbb{Z}_2$, the swap symmetry $(w_1, w_2) \mapsto (w_2, w_1)$.

Singularity: The loss landscape $L(w_1, w_2)$ is invariant under this swap. Any minimum (w_1^*, w_2^*) has a "twin" (w_2^*, w_1^*) at the same loss value. Along the diagonal $w_1 = w_2$, this swap is trivial, creating a line of equivalent minima (a singularity).

Now suppose the agent refines its semantics to distinguish the two input channels:

Refined partition: $\Pi_{\text{fine}} = \{\{w_1\}, \{w_2\}\}$—each weight in its own block.

Broken symmetry: $\text{Aut}(\Pi_{\text{fine}}) = \{e\}$ (trivial group). The swap is no longer a symmetry.

Changed singularity: If the refined semantics are implemented (e.g., via different regularization, task structure, or architectural constraints that treat w_1 and w_2 differently), the loss landscape loses its swap invariance. The diagonal is no longer special; minima become isolated.

This illustrates how semantic refinement (distinguishing the two channels) breaks a symmetry and changes the singularity structure.

2.3 Symmetry and RLCT: an SLT connection

The previous example showed symmetry creating a singularity qualitatively. We now make the RLCT connection explicit.

Example 2.2 (RLCT and symmetry breaking). Consider a loss function $K(w_1, w_2) = (w_1 - w_2)^2$ near the minimum set $\{w_1 = w_2\}$ (a line). This has:

- Swap symmetry: $K(w_1, w_2) = K(w_2, w_1)$.

- RLCT $\lambda = 1/2$ (the minimum is a 1-dimensional manifold in 2-dimensional space; standard SLT calculation).

Now break the symmetry by adding a term that distinguishes the weights:

$$K_\epsilon(w_1, w_2) = (w_1 - w_2)^2 + \epsilon w_1^2.$$

For $\epsilon > 0$:

- The swap symmetry is broken.
- The minimum is now isolated at $(0, 0)$.
- RLCT $\lambda = 1$ (isolated non-degenerate minimum in 2D).

The RLCT jumped from 1/2 to 1 when the symmetry was broken. In SLT, lower RLCT indicates more severe singularity (slower posterior concentration, higher model complexity penalty). Symmetry breaking increased RLCT by removing the flat direction.

Remark 2.3 (General principle). More generally, when a k-dimensional group G acts on parameter space leaving the loss invariant, minima typically lie on G-orbits, creating at least $\dim(G)$ flat directions. Breaking the G-symmetry removes these flat directions, generically increasing RLCT. This is the mechanism by which partition refinement (which shrinks symmetry groups) can increase RLCT and change singularity type.

3 Formalizing Semantics, Partitions, and Faithfulness

We now make the conceptual chain precise.

3.1 Semantic states via probes

Definition 3.1 (Semantic probe family and semantic state)**.** Let $\mathcal{Q}$ be a fixed set of *semantic probes*—queries or tests interrogating goal-relevant behavior. A *semantic state* is a response function $s : \mathcal{Q} \to \mathcal{Y}$, where $\mathcal{Y}$ is a response set.

Two semantic states s, s' are *semantically equivalent*, written $s \sim_{\text{sem}} s'$, if $s(q) = s'(q)$ for all $q \in \mathcal{Q}$.

A *semantic goal equilibrium* is an equivalence class $e = [s]_{\sim_{\text{sem}}}$.

3.2 From semantic states to partitions: the key bridge

The following definition makes explicit how semantics induce partitions on underlying degrees of freedom.

Definition 3.2 (Semantic-partition map)**.** Let U be a finite set of *underlying elements* (parameters, features, states, or other degrees of freedom, depending on architecture). A *semantic-partition map* is a function

$$\Phi : \{\text{semantic states}\} \to \{\text{partitions of } U\},$$

where $\Phi(s) = \Pi_s$ is defined by: $u \sim_{\Pi_s} u'$ iff u and u' are *not distinguished* by the agent's responses under semantic state s.

Concretely, if probes in $\mathcal{Q}$ can be evaluated on elements of U (or on pairs, or on functions thereof), then:

$$u \sim_{\Pi_s} u' \iff \forall q \in \mathcal{Q} : s(q)(u) = s(q)(u'),$$

where $s(q)(u)$ denotes the probe response at element u.

Remark 3.3 (Architecture-dependence). The set U and the precise form of Φ depend on the architecture:

- For a neural network, U might be edges or hidden units, and $\Phi(s)$ partitions them by functional role under semantics s.
- For a compression scheme, U might be information atoms, partitioned by which feature "owns" them.
- For a logical reasoner, U might be possible worlds, partitioned by epistemic indistinguishability.

The instantiations in Section 5 make these choices explicit.

Definition 3.4 (Semantic coarsening and refinement)**.** Semantic state s' is a *coarsening* of s if $\Phi(s) \preceq \Phi(s')$, i.e., every block of Π_s is contained in some block of $\Pi_{s'}$. Equivalently, s' makes fewer distinctions than s.

Semantic state s' is a *refinement* of s if $\Phi(s') \preceq \Phi(s)$.

3.3 Singularity types in SLT

Definition 3.5 (Singularity-type equivalence)**.** Let $K : \Theta \subseteq \mathbb{R}^d \to \mathbb{R}_{\geq 0}$ be real-analytic with $K(\theta^*) = 0$. Two such functions K, K' with zeros at $\theta^*, \theta^{*\prime}$ have *equivalent singularity types* if there exist neighborhoods U, U', an analytic diffeomorphism $\varphi : U \to U'$ with $\varphi(\theta^*) = \theta^{*\prime}$, and a positive analytic function $u : U \to (0, \infty)$ such that

$$K(\theta) = u(\theta) \cdot K'(\varphi(\theta)) \quad \text{for all } \theta \in U.$$

Equivalent singularity types share all SLT invariants (RLCT λ, multiplicity m, zeta-function data).

3.4 The faithfulness assumption

Definition 3.6 (Singularity-type map)**.** Let $\mathcal{E}$ be the set of semantic goal equilibria. For each $e \in \mathcal{E}$, let $K_e : \Theta_e \to \mathbb{R}_{\geq 0}$ be the analytic loss governing learning in equilibrium e. Define $\mathrm{SingType}(e)$ as the singularity type of K_e at a designated representative minimum.

Assumption A1 (Semantic–Singularity Faithfulness)*.* The singularity-type map is injective:

$$e \neq e' \quad \Longrightarrow \quad \mathrm{SingType}(e) \ncong \mathrm{SingType}(e').$$

Remark 3.7 (Status of faithfulness)*.* Assumption A1 is a structural hypothesis, not a general theorem. The contribution of this paper is to identify *when* it is plausible via the distinction-symmetry mechanism. Section 6 states a genericity conjecture.

3.5 The detection theorem

Theorem 3.8 (Semantic change detection)**.** *Under Assumption A1, if a monitoring procedure detects* $\mathrm{SingType}(e(t_0)) \ncong \mathrm{SingType}(e(t_1))$*, then* $e(t_0) \neq e(t_1)$*: a semantic change has occurred.*

Proof. Contrapositive of injectivity. □

Remark 3.9 (Practical detection)*.* In practice, one estimates coarse invariants like RLCT. A change in RLCT implies a singularity-type change, but non-equivalent types may share RLCT. Thus some semantic changes could go undetected by RLCT alone.

4 The Partition Template Theorem

We now prove the mathematical core: partitions control symmetries monotonically.

4.1 Definitions

Definition 4.1 (Partition and coarsening order)**.** Let U be finite. A *partition* $\Pi = \{B_1, \dots, B_m\}$ of U is a collection of nonempty, pairwise-disjoint blocks covering U.

We write $\Pi \preceq \Pi'$ (read: "Π' is a coarsening of Π" or "Π refines Π'") if every block of Π is contained in some block of Π'. Equivalently, Π' is obtained by merging blocks of Π.

Definition 4.2 (Automorphism group)**.** For partition $\Pi = \{B_1, \dots, B_m\}$, define

$$\mathrm{Aut}(\Pi) := \prod_{j=1}^{m} \mathrm{Sym}(B_j).$$

Elements of $\mathrm{Aut}(\Pi)$ are permutations of U preserving each block setwise. Intuitively, $\mathrm{Aut}(\Pi)$ consists of permutations "invisible" to any function depending only on block membership.

Definition 4.3 (Overlap functionals)**.** For partition $\Pi = \{B_j\}$ and weights $w : U \to [0, \infty)$:

$$\mathrm{Ov}_{\mathrm{node}}(\Pi; w) := \sum_j \left(\sum_{u \in B_j} w(u) \right)^2 = \|(\text{block masses})\|_2^2.$$

When $w = p$ is a probability distribution, this equals the *collision probability* $\sum_j p(B_j)^2$.

For symmetric weights $W : U \times U \to [0, \infty)$:

$$\mathrm{Ov}_{\mathrm{pair}}(\Pi; W) := \sum_j \sum_{u,v \in B_j} W(u, v).$$

4.2 The template theorem

Lemma 4.4 (Single-merge effect)**.** *Merging two blocks C, D with masses $S_C = \sum_{u \in C} w(u)$ and $S_D = \sum_{u \in D} w(u)$:*

1. *Increases $|\mathrm{Aut}(\Pi)|$ by factor $\binom{|C|+|D|}{|C|}$ (from new cross-block permutations).*
2. *Increases $\mathrm{Ov}_{\mathrm{node}}$ by $2S_C S_D \geq 0$ (strictly positive if both masses positive).*

Proof. (1) $|\mathrm{Sym}(C)| \cdot |\mathrm{Sym}(D)| = |C|! \cdot |D|!$ becomes $|\mathrm{Sym}(C \cup D)| = (|C| + |D|)!$, with ratio $\binom{|C|+|D|}{|C|}$.

(2) $(S_C + S_D)^2 - (S_C^2 + S_D^2) = 2S_C S_D$. □

Theorem 4.5 (Coarsening increases symmetry and overlap)**.** *Let* $\Pi \preceq \Pi'$ *(i.e.,* Π' *coarsens* Π*). Then:*

1. ***(Monotone symmetry)*** $\mathrm{Aut}(\Pi) \subseteq \mathrm{Aut}(\Pi')$, *with strict inclusion if* $\Pi' \neq \Pi$.

2. ***(Monotone node-overlap)*** $\mathrm{Ov}_{\mathrm{node}}(\Pi; w) \leq \mathrm{Ov}_{\mathrm{node}}(\Pi'; w)$, *strictly if a merge combines positive-mass blocks.*

3. ***(Monotone pair-overlap)*** $\mathrm{Ov}_{\mathrm{pair}}(\Pi; W) \leq \mathrm{Ov}_{\mathrm{pair}}(\Pi'; W)$, *strictly if a positive-weight pair becomes newly co-blocked.*

Proof. Each Π'-block is a union of Π-blocks. Any Π-preserving permutation preserves each union, hence lies in $\mathrm{Aut}(\Pi')$. Strictness: a nontrivial merge permits cross-block permutations absent in $\mathrm{Aut}(\Pi)$.

Parts (2) and (3) follow by applying Lemma 4.4 iteratively and noting that coarsening only adds co-blocked pairs. □

Corollary 4.6 (Refinement decreases symmetry and overlap)**.** *If* Π' *strictly refines* Π, *then* $\mathrm{Aut}(\Pi') \subsetneq \mathrm{Aut}(\Pi)$ *and overlaps decrease (for appropriate weights).*

Remark 4.7 (The mechanism in summary). Theorem 4.5 plus the SLT connection (Example 2.2) yields the full chain: semantic coarsening → partition coarsening → larger symmetry group → more flat directions → lower RLCT / more severe singularity. Refinement reverses each arrow.

5 Three Instantiations

Each instantiation specifies U, the semantic-partition map Φ, and how coarsening/refinement arise operationally.

5.1 Causal-Coding Neural Networks

5.1.1 Setting

Causal-coding architectures use gates or masks to control which edges are active for which contexts/tasks. A key challenge is parameter entanglement:

when contexts share edges indiscriminately, many edges become interchangeable, creating redundancy and flat loss landscapes.

5.1.2 Semantic-partition map

Underlying set: $U = E$ (the set of edges/units).

Semantic state: The gating pattern $D = (g_e^c)_{e \in E, c \in C}$ encodes which edges are active in which contexts. This implements the agent's semantic distinctions over contexts: two edges are *semantically indistinguishable under* D iff they have identical context-profiles $\chi_D(e) := (g_e^c)_{c \in C}$.

Partition: $\Phi(D) = \Pi(D)$, where $e \sim_{\Pi(D)} e'$ iff $\chi_D(e) = \chi_D(e')$.

Interpretation: Semantic coarsening (treating more contexts as equivalent, or using edges less discriminatively) corresponds to partition coarsening. Refining semantic distinctions between contexts corresponds to partition refinement.

5.1.3 Symmetry and overlap

The automorphism group is $\mathrm{Aut}(\Pi(D)) = \prod_{B \in \Pi(D)} \mathrm{Sym}(B)$: edges with identical profiles are interchangeable (permutation symmetry in the loss).

For overlap, define sharedness weight $w_D(e) := |\{(c, c') : g_e^c = g_e^{c'} = 1\}|$ and $\mathrm{Ov}_{\mathrm{CC}}(D) := \mathrm{Ov}_{\mathrm{node}}(\Pi(D); w_D)$.

5.1.4 Coarsening theorem

Definition 5.1 (Partition coarsening for gating)**.** Gating pattern D' is a *coarsening* of D if $\Pi(D) \preceq \Pi(D')$, i.e., whenever $\chi_D(e) = \chi_D(e')$, we also have $\chi_{D'}(e) = \chi_{D'}(e')$.

Equivalently: the equivalence relation induced by D is contained in that induced by D'.

Remark 5.2 (On gate-wise conditions)*.* Note that entry-wise inequality $g_e^{c'} \geq g_e^c$ is *not* sufficient for partition coarsening. Activating an edge in additional contexts can split equivalence classes (if it creates a new profile). Definition 5.1 works directly at the partition level.

Corollary 5.3 (Causal-coding coarsening)**.** *If D' coarsens D (Definition 5.1), then*

$$\mathrm{Aut}(\Pi(D)) \subseteq \mathrm{Aut}(\Pi(D')), \qquad \mathrm{Ov}_{\mathrm{CC}}(D) \leq \mathrm{Ov}_{\mathrm{CC}}(D').$$

Proof. Apply Theorem 4.5. $\square$

5.1.5 Refinement via regularization

Proposition 5.4 (Overlap penalties add curvature)**.** *Let $J(\theta)$ be the base objective and $R(\theta) \geq 0$ an overlap penalty increasing with* $\mathrm{Ov}_{\mathrm{CC}}$*. Consider $J_\lambda(\theta) := J(\theta) + \lambda R(\theta)$.*

Assume: (i) $\arg\min J$ *is nonempty and compact; (ii) R is nonconstant on* $\arg\min J$*; (iii) $\nabla^2 R \succeq \nu I$ on a subspace V of "entanglement directions" at minimizers.*

Then minimizers of J_λ have strictly lower R than the worst J-minimizer, and $\nabla^2 J_\lambda$ has curvature $\geq \lambda\nu$ along V.

Proof. Adding λR selects for lower R among J-minimizers (standard regularization argument). Curvature follows from $\nabla^2 J_\lambda = \nabla^2 J + \lambda \nabla^2 R$. □

5.2 Incremental Compression

5.2.1 Setting

Incremental compression builds representations feature-by-feature. A pathology is level entanglement: early features absorb structure belonging to later features. If early features are frozen, this creates redundancy.

5.2.2 Semantic-partition map

Underlying set: $U =$ information atoms (micro-regularities in the source).

Semantic state: A representation F (the current features) determines which atoms are "owned" by which feature/level. The semantic distinctions are about *which structural level handles which patterns*: atoms owned by the same feature are treated as "same-level structure."

Partition: $\Phi(F) = \Pi(F)$, the ownership partition where atoms assigned to the same feature are co-blocked.

Interpretation: Semantic coarsening occurs when ownership becomes less specific (more atoms lumped under one owner)—this is level entanglement. Semantic refinement occurs when ownership becomes more specific (atoms reassigned to their proper levels)—this is disentanglement.

5.2.3 Coarsening theorem

Corollary 5.5 (Compression coarsening)**.** *If* $\Pi(F) \preceq \Pi(F')$*, then* $\mathrm{Aut}(\Pi(F)) \subseteq \mathrm{Aut}(\Pi(F'))$ *and* $\mathrm{Ov}_{\mathrm{IC}}(F) \leq \mathrm{Ov}_{\mathrm{IC}}(F')$*.*

5.2.4 Refinement via refitting

Definition 5.6 (Refit operator)**.** A *refit step* re-optimizes all features jointly when a new feature is added: $\mathrm{Refit}(F) \in \arg\min_G L(G)$ for description-length objective L.

Proposition 5.7 (Refitting tends to refine (heuristic))**.** *Assume: (i) the source has latent hierarchical structure; (ii) mixed-level ownership strictly increases L (formally: for any F and any nontrivial merge of blocks in* $\Pi(F)$ *creating mixed-level ownership, L strictly increases).*

Then refitting yields $\Pi(F_{\mathrm{refit}}) \preceq \Pi(F_{\mathrm{freeze}})$ *and* $\mathrm{Ov}_{\mathrm{IC}}(F_{\mathrm{refit}}) \leq \mathrm{Ov}_{\mathrm{IC}}(F_{\mathrm{freeze}})$.

Remark 5.8 (Status of this claim). Proposition 5.7 is stated as a heuristic: the conclusion follows from the assumptions, but verifying assumption (ii) requires analyzing the specific objective L and source structure. The key point is that *if* L penalizes mixed ownership sufficiently, *then* joint optimization refines the partition.

5.3 PLN Inference (Logic Quantale View)

5.3.1 Setting

Probabilistic logical reasoning maintains beliefs over possible worlds. A knowledge state induces indistinguishability: worlds consistent with current evidence are grouped together.

5.3.2 Semantic-partition map

Underlying set: $U = W$ (possible worlds).

Semantic state: The agent's knowledge/evidence K determines which worlds can be distinguished. Two worlds are semantically equivalent iff current knowledge cannot tell them apart.

Partition: $\Phi(K) = \Pi_K$, where $w \sim_{\Pi_K} w'$ iff w and w' are epistemically indistinguishable given K.

Interpretation: This is the most direct case: semantics *are* the epistemic distinctions. Coarsening (forgetting evidence) merges indistinguishability classes. Refinement (gaining evidence) splits them.

5.3.3 Overlap as collision probability

Let p be a prior on W. Define $\mathrm{Ov}_{\mathrm{PLN}}(\Pi) := \sum_j p(B_j)^2$, the probability that two independent draws land in the same indistinguishability class.

5.3.4 Template instantiation

Corollary 5.9 (PLN coarsening and refinement)**.** *If* $\Pi_K \preceq \Pi_{K'}$ *(less evidence), then* $\mathrm{Aut}(\Pi_K) \subseteq \mathrm{Aut}(\Pi_{K'})$ *and* $\mathrm{Ov}_{\mathrm{PLN}}(\Pi_K) \leq \mathrm{Ov}_{\mathrm{PLN}}(\Pi_{K'})$*.*

Refinement (more evidence) reverses these inequalities.

Remark 5.10 (Logic quantale interpretation)*.* In the quantale framing, predicates partition W into satisfying/non-satisfying worlds. Adding predicates intersects partition blocks, monotonically decreasing collision probability. "Making more distinctions" rigorously reduces epistemic redundancy.

6 Discussion

6.1 When is faithfulness justified?

The distinction-symmetry-singularity chain provides sufficient conditions for Assumption A1:

1. **Semantic-partition coupling:** The semantic-partition map Φ (Definition 3.2) is well-defined and injective on semantic equivalence classes.

2. **Partition-symmetry coupling:** The partition $\Phi(s)$ controls symmetries of the learning objective via parameter equivalences.

3. **Symmetry-singularity coupling:** Different symmetry structures yield different singularity types (generically true for non-identifiable models in general position).

Conjecture 6.1 (Genericity of faithfulness)**.** *For architectures where semantics are realized via partitions as in Section 4, and where the parameterization is in* general position *(no accidental algebraic relations beyond those forced by symmetries), distinct semantic equilibria almost surely lead to distinct singularity types.*

This conjecture is not proved here, but it sharpens the claim: faithfulness should hold "generically" when the three couplings are tight, and fail when they are loose (e.g., when semantics are encoded in discrete choices orthogonal to continuous parameter symmetries).

6.2 Partial faithfulness

In practice, full injectivity may be too strong. A useful relaxation:

Definition 6.2 (Partial faithfulness)**.** Let $\approx$ be a coarse equivalence on semantic equilibria (e.g., "same goal family"). The singularity-type map is *partially faithful mod* $\approx$ if

$$e \not\approx e' \implies \mathrm{SingType}(e) \ncong \mathrm{SingType}(e').$$

This allows singularity types to collide within a goal family while still detecting cross-family drift.

6.3 Limitations

Remark 6.3 (Coarse invariants)*.* RLCT and multiplicity may not distinguish all non-equivalent singularity types. Comprehensive detection may require richer invariants or complementary semantic probes.

Remark 6.4 (Architecture-specificity)*.* The template theorem is universal, but instantiation requires identifying U, Φ, and coarsening/refinement operations for each architecture.

Remark 6.5 (Detection vs. prevention)*.* This paper addresses detection only. Preventing goal drift requires additional mechanisms beyond monitoring.

6.4 Failure modes

The faithfulness assumption can fail when:

- Semantics are encoded in discrete architectural choices (e.g., "which module is active") rather than continuous parameter structure.
- The parameterization has accidental symmetries unrelated to semantics.
- Multiple semantic equilibria happen to share loss landscapes up to analytic equivalence (a non-generic but possible coincidence).

7 Conclusion

We developed a mechanism linking semantic changes to geometric changes detectable by SLT:

$$\textbf{Semantic distinctions} \xrightarrow{\Phi} \textbf{Partitions} \xrightarrow{\text{Thm 4.5}} \textbf{Symmetry groups} \xrightarrow{\text{SLT}} \textbf{Singularity types}$$

The template theorem (Theorem 4.5) establishes that coarsening distinctions monotonically increases symmetries, while refinement breaks them. Combined with the SLT principle that symmetries generate singularities (Example 2.2), this provides a concrete, falsifiable pathway from semantic change to geometric change.

We formalized the faithfulness assumption, identified sufficient conditions via the distinction-symmetry mechanism, stated a genericity conjecture, and demonstrated the template's scope across three AI architectures.

This work provides theoretical grounding for using SLT diagnostics in goal stability monitoring, by clarifying when geometric changes reliably indicate semantic changes.

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